

Animal Database Design

Caden Raleigh

ISIN-325-001

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Overview:

A relational database stores information in separate tables to avoid repeating the same data many times. Each table stores a specific subject, such as Animal, Species, Habitat, Diet, and Location. Rows inside these tables represent real-world records. Primary Keys (PK) are used to make each record unique. Foreign Keys (FK) connect one table to another so that data stays consistent and accurate.

Normalization is also important because it prevents duplicate values and limits problems when updating or deleting information. For example, instead of putting habitat names directly inside the Animal table for every record, the database uses HabitatID as a foreign key. This points to the Habitat table, which stores the real description. This reduces errors and avoids repeating the same description many times.

This database design results in better organization, easier maintenance, and improved data quality.

Standards:

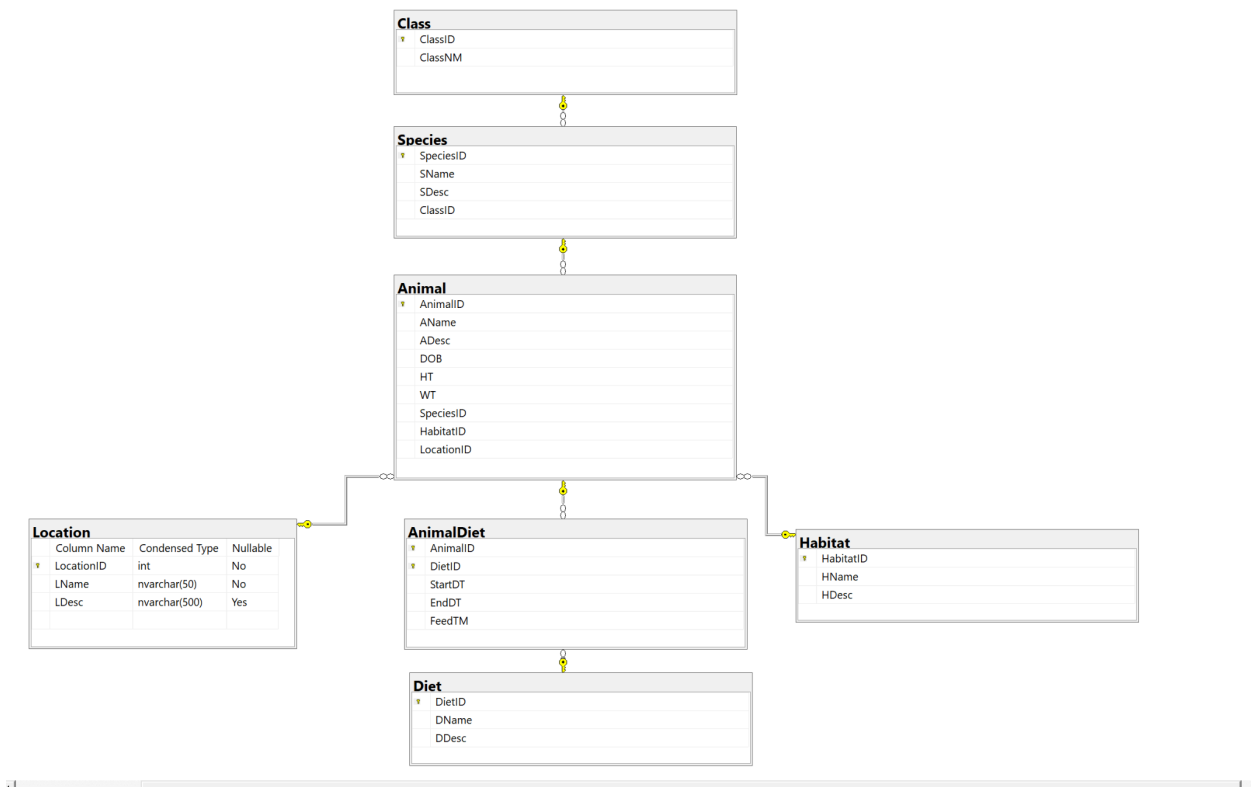
Database Naming Standards: Consistent naming helps others understand the structure of the database. Names should be short, meaningful, and follow a clear pattern. For example, this database uses names like AnimalID, SpeciesID, HabitatID, DietID, and LocationID which clearly show the purpose of each column (Database Naming Standards).

Security Standards: Security standards help protect data from unauthorized access. PKs and FKs support security by controlling how data is linked and accessed. Security standards also include controlling who gets specific access, use of stored procedures, and setting the correct security mode on servers (Security Standards).

Backup/Recovery Standards: Backups and recovery procedures are critical to protect important data. The difficult part is not just taking the backup, but knowing where it is and how to perform the recovery correctly. Practicing recovery helps avoid data loss and prepares for emergencies (Backup/Recovery Standards).

Database Illustration:

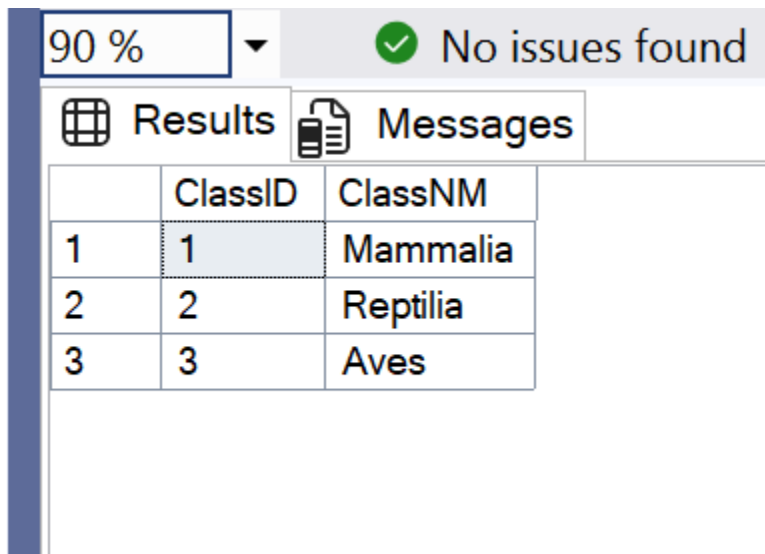
DB2 Caden Raleigh 11/2/2025



Proof of Population:

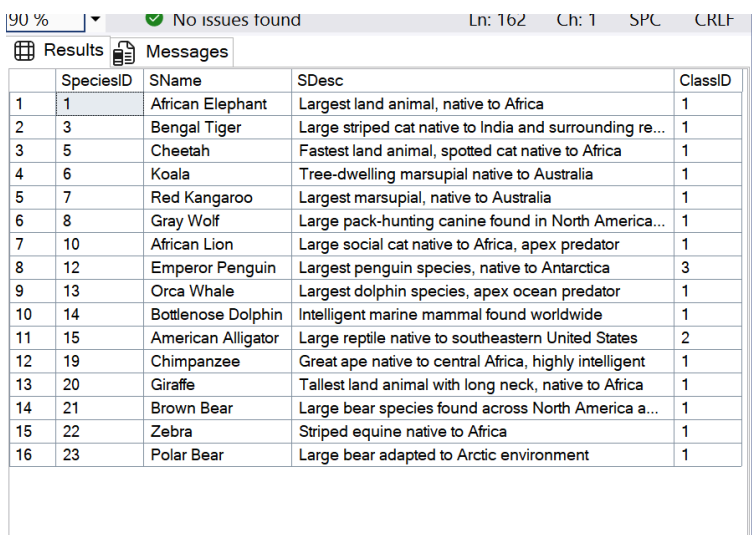
Below are screenshots proving that each table has been populated with actual records.
Each screenshot shows the results of a SELECT statement confirming that rows exist.

Class Table (SELECT * FROM Class):



	ClassID	ClassNM
1	1	Mammalia
2	2	Reptilia
3	3	Aves

Species Table (SELECT * FROM Species):



	SpeciesID	SName	SDesc	ClassID
1	1	African Elephant	Largest land animal, native to Africa	1
2	3	Bengal Tiger	Large striped cat native to India and surrounding re...	1
3	5	Cheetah	Fastest land animal, spotted cat native to Africa	1
4	6	Koala	Tree-dwelling marsupial native to Australia	1
5	7	Red Kangaroo	Largest marsupial, native to Australia	1
6	8	Gray Wolf	Large pack-hunting canine found in North America...	1
7	10	African Lion	Large social cat native to Africa, apex predator	1
8	12	Emperor Penguin	Largest penguin species, native to Antarctica	3
9	13	Orca Whale	Largest dolphin species, apex ocean predator	1
10	14	Bottlenose Dolphin	Intelligent marine mammal found worldwide	1
11	15	American Alligator	Large reptile native to southeastern United States	2
12	19	Chimpanzee	Great ape native to central Africa, highly intelligent	1
13	20	Giraffe	Tallest land animal with long neck, native to Africa	1
14	21	Brown Bear	Large bear species found across North America a...	1
15	22	Zebra	Striped equine native to Africa	1
16	23	Polar Bear	Large bear adapted to Arctic environment	1

Habitat Table (SELECT * FROM Habitat):

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Results		Messages	
	HabitatID	HName	HDesc
1	1	Savanna	A grassy plain with scattered trees, found in tropical a...
2	2	Rainforest	A dense, warm, and wet forest with high biodiversity, f...
3	3	Desert	A dry region with sparse vegetation and extreme tem...
4	4	Ocean	The largest habitat on Earth, home to diverse marine l...
5	5	Arctic Tundra	A cold, treeless region with permafrost and low-growi...
6	6	Freshwater	Includes rivers, lakes, and wetlands, supporting fish, a...
7	7	Grassland	Wide open areas dominated by grasses, often suppor...
8	8	Forest	A large area covered chiefly with trees and undergro...
9	9	Mountain	High-altitude regions with rocky terrain and unique ec...
10	10	Urban	Human-made environments such as cities and towns ...

Location Table (SELECT * FROM Location):

Results		Messages	
	LocationID	LName	LDesc
1	1	North Enclosure	Primary exhibit area A
2	2	South Enclosure	Primary exhibit area B
3	3	East Enclosure	Primary exhibit area C
4	4	West Enclosure	Primary exhibit area D
5	5	Aquatics Bay	Marine exhibits and pools

Diet Table (SELECT * FROM Diet):

	Results	Messages	
	DietID	DName	DDesc
1	1	Carnivore Diet	Composed of raw meats such as beef, chicken, and...
2	2	Herbivore Diet	A high-fiber blend of grasses, hay, and leafy greens ...
3	3	Omnivore Diet	A varied combination of meats, grains, vegetables, a...
4	4	Piscivore Diet	Contains fresh or frozen fish and shellfish, offering hi...
5	5	Insectivore Diet	Includes live crickets, mealworms, and beetles enric...
6	6	Frugivore Diet	A mix of tropical fruits like bananas, mangoes, and p...
7	7	Granivore Diet	Formulated from a mix of seeds, nuts, and grains pr...
8	8	Nectarivore Diet	A sweetened nectar formula with fruit puree and ami...
9	9	Folivore Diet	Primarily composed of eucalyptus leaves, bamboo, ...
10	10	Detritivore Diet	Includes decaying plant matter, fungi, and compost t...
11	11	Piscivorous Bird Diet	A seafood-based diet featuring small fish and crusta...
12	12	Reptile Carnivore Diet	Raw meat and whole prey items such as mice and c...
13	13	Amphibian Diet	A moist, insect-heavy blend including worms and fl...
14	14	Marine Mammal Diet	Consists of fresh fish, squid, and vitamin supplement...
15	15	Avian Herbivore Diet	A mix of leafy greens, fruits, and seeds suited for par...
16	16	Primate Diet	Balanced portions of fruits, vegetables, nuts, and pro...
17	17	Rodent Diet	A combination of grains, seeds, and fresh vegetable...
18	18	Ungulate Diet	Specialized hay and mineral-enriched grain mix for ...
19	19	Arthropod Diet	Comprised of decaying leaves, fruit scraps, and calc...
20	20	Piscivore Reptile Diet	High-protein fish and crustacean mix developed for ...
21	21	Seed and Fruit Mix Diet	Combination of seeds, nuts, and fruit pieces design...
22	22	Crustacean Diet	A diet of algae, fish flakes, and plant matter suitable f...
23	23	Nocturnal Mammal Diet	A mix of fruits, insects, and small prey designed for ...
24	24	Desert Herbivore Diet	Dried grasses, succulents, and fiber-rich feed create...
25	25	Rainforest Diet	A rich variety of fruits, insects, and leaves formulated...

Animal Table (SELECT * FROM Animal):

	AnimalID	AName	ADesc	DOB	HT	WT	SpeciesID	H
1	1	Simba	Male African Lion	2018-05-12	1.20	190.50	10	1
2	2	Nala	Female African Lion	2019-07-08	1.10	165.00	10	1
3	3	Dumbo	African Elephant calf	2021-04-15	2.30	450.00	1	1
4	4	Ellie	Adult African Elephant	2010-03-09	3.20	5200.00	1	1
5	5	Stripe	Male Bengal Tiger	2016-11-22	1.00	220.00	3	2
6	6	Tara	Female Bengal Tiger	2018-06-30	0.95	200.00	3	2
7	7	Splash	Bottlenose Dolphin	2020-02-10	2.70	180.00	14	4
8	8	Fluke	Young Bottlenose Dolphin	2023-08-25	1.90	95.00	14	4
9	9	Rocky	Brown Bear	2015-09-12	1.30	300.00	21	8
10	10	Bella	Female Brown Bear	2017-05-18	1.10	250.00	21	8
11	11	Zara	Female Zebra	2017-11-02	1.40	350.00	22	7
12	12	Marty	Male Zebra	2018-03-14	1.50	380.00	22	7
13	13	Milo	Chimpanzee	2014-08-10	0.90	45.00	19	6
14	14	Bobo	Male Chimpanzee	2016-02-25	0.95	50.00	19	6
15	15	Luna	Female Polar Bear	2013-12-18	1.80	450.00	23	5
16	16	Blizzard	Male Polar Bear	2012-10-03	1.90	500.00	23	5
17	17	Echo	Male Gray Wolf	2015-07-25	0.95	55.00	8	7
18	18	Winston	Alpha Gray Wolf	2014-03-19	1.00	60.00	8	7
19	19	Marley	Male Koala	2019-04-12	0.60	14.00	6	8
20	20	Coco	Female Koala	2020-06-15	0.58	13.50	6	8
21	21	Killer	Orca Whale	2011-05-01	6.00	5400.00	13	4
22	22	Shadow	Female Orca Whale	2012-03-10	5.80	5100.00	13	4
23	23	Penny	Penguin	2020-11-22	0.75	5.50	12	5
24	24	Tux	Male Emperor Penguin	2019-02-18	0.80	6.00	12	5
25	25	Nova	Giraffe	2017-01-09	4.00	800.00	20	7
26	26	Fiona	Female Giraffe	2018-04-25	3.90	750.00	20	7
27	27	Ash	Male Red Kangaroo	2020-08-30	2.20	80.00	7	8
28	28	Ruby	Female Red Kangaroo	2021-01-15	2.00	75.00	7	8
29	29	Leo	Lioness	2016-06-30	1.00	150.00	10	1
30	30	Tori	Female Bengal Tiger	2015-12-09	1.10	190.00	3	2
31	31	Cleo	Lioness	2018-03-04	1.05	160.00	10	1
32	32	Toby	Male American Alligator	2014-07-07	4.00	450.00	15	2
33	33	Sally	Female American Alligator	2016-10-17	3.80	400.00	15	2
34	34	Theo	Male Cheetah	2017-11-30	1.20	65.00	5	7
35	35	Luna	Female Cheetah	2018-02-14	1.10	60.00	5	7
36	36	Rex	Male Giraffe	2019-04-03	4.10	850.00	20	7
37	37	Sasha	Female Giraffe	2017-06-23	3.90	800.00	20	7
38	38	Ace	African Elephant Bull	2010-02-25	3.30	5200.00	1	1
39	39	Olive	Female African Elephant	2013-09-09	3.00	4500.00	1	1
40	40	Spike	Male Zebra	2015-12-30	1.55	400.00	22	7
41	41	Koda	Male Grizzly Bear	2016-05-14	1.50	350.00	21	8
42	42	Raven	Female Grizzly Bear	2017-03-20	1.40	320.00	21	8
43	43	Toby	Male Polar Bear	2014-08-05	1.85	480.00	23	5
44	44	Nina	Female Polar Bear	2015-02-18	1.80	450.00	23	5
45	45	Blake	Male Giraffe	2019-01-11	4.00	700.00	20	7
46	46	Sophie	Female Giraffe	2018-06-25	3.90	680.00	20	7
47	47	Mia	Female Bengal Tiger	2017-09-12	1.00	200.00	3	2
48	48	Ace	Male Bengal Tiger	2016-03-22	1.10	230.00	3	2
49	49	Hunter	Male African Lion	2014-10-15	1.20	210.00	10	1
50	50	Luna	Female African Lion	2016-02-10	1.10	180.00	10	1
51	51	Theo	Male Cheetah	2017-05-03	1.10	60.00	5	7
52	52	Jasmine	Female Cheetah	2018-04-29	1.00	55.00	5	7
53	53	Eliza	Female Red Kangaroo	2020-03-02	2.00	70.00	7	8
54	54	Jack	Male Red Kangaroo	2021-02-18	2.10	75.00	7	8
55	55	Max	Male Zebra	2018-12-13	1.50	380.00	22	7
56	56	Zara	Female Zebra	2017-09-22	1.45	350.00	22	7

AnimalDiet Table (SELECT * FROM AnimalDiet):

	AnimalID	DietID	StartDT	EndDT	FeedTM
1	1	1	NULL	NULL	NULL
2	1	2	NULL	NULL	NULL
3	2	1	NULL	NULL	NULL
4	3	2	NULL	NULL	NULL
5	4	3	NULL	NULL	NULL
6	5	1	NULL	NULL	NULL
7	6	1	NULL	NULL	NULL
8	7	14	NULL	NULL	NULL
9	8	14	NULL	NULL	NULL
10	9	3	NULL	NULL	NULL
11	10	3	NULL	NULL	NULL
12	11	2	NULL	NULL	NULL
13	12	2	NULL	NULL	NULL
14	13	16	NULL	NULL	NULL
15	14	16	NULL	NULL	NULL
16	15	3	NULL	NULL	NULL
17	16	3	NULL	NULL	NULL
18	17	1	NULL	NULL	NULL
19	18	1	NULL	NULL	NULL
20	19	9	NULL	NULL	NULL
21	20	9	NULL	NULL	NULL
22	21	14	NULL	NULL	NULL
23	22	14	NULL	NULL	NULL
24	23	11	NULL	NULL	NULL
25	24	11	NULL	NULL	NULL
26	25	2	NULL	NULL	NULL
27	26	2	NULL	NULL	NULL
28	27	9	NULL	NULL	NULL
29	28	9	NULL	NULL	NULL
30	29	1	NULL	NULL	NULL
31	30	1	NULL	NULL	NULL
32	31	1	NULL	NULL	NULL
33	32	1	NULL	NULL	NULL
34	33	1	NULL	NULL	NULL

Row Count Summary:

Results

Messages

	TableName	Rows
1	Class	3
2	Species	16
3	Habitat	10
4	Location	5
5	Diet	25
6	Animal	68
7	AnimalDiet	73

Results

Messages

ClassID	ClassNM	
1	Mammalia	
2	Reptilia	
3	Aves	

SpeciesID	SName	SDesc	ClassID
1	African Elephant	Largest land animal, native to Africa	1
2	Bengal Tiger	Large striped cat native to India and surroundi...	1
3	Cheetah	Fastest land animal, spotted cat native to Africa	1
4	Koala	Tree-dwelling marsupial native to Australia	1
5	Red Kangaroo	Largest marsupial, native to Australia	1

HabitatID	HName	HDesc
1	Savanna	A grassy plain with scattered trees, found in tropi...
2	Rainforest	A dense, warm, and wet forest with high biodiver...
3	Desert	A dry region with sparse vegetation and extreme ...
4	Ocean	The largest habitat on Earth, home to diverse ma...
5	Arctic Tundra	A cold, treeless region with permafrost and low-g...

LocationID	LName	LDesc
1	North Enclosure	Primary exhibit area A
2	South Enclosure	Primary exhibit area B
3	East Enclosure	Primary exhibit area C
4	West Enclosure	Primary exhibit area D
5	Aquatics Bay	Marine exhibits and ...

DietID	DName	DDesc
1	Carnivore Diet	Composed of raw meats such as beef, chicken, an...
2	Herbivore Diet	A high-fiber blend of grasses, hay, and leafy green...
3	Omnivore Diet	A varied combination of meats, grains, vegetables, ...
4	Piscivore Diet	Contains fresh or frozen fish and shellfish, offering h...
5	Insectivore ...	Includes live crickets, mealworms, and beetles enri...

AnimalID	AName	ADesc	DOB	HT	WT	SpeciesID	HabitatID	LocationID	
1	1	Simba	Male African Lion	2018-05-12	1.20	190.50	10	1	1
2	2	Nala	Female African...	2019-07-08	1.10	165.00	10	1	1
3	3	Dumbo	African Elepha...	2021-04-15	2.30	450.00	1	1	2
4	4	Elie	Adult African El...	2010-03-09	3.20	5200...	1	1	2
5	5	Stripe	Male Bengal Ti...	2016-11-22	1.00	220.00	3	2	3
6	6	Tara	Female Bengal...	2018-06-30	0.95	200.00	3	2	3
7	7	Splash	Bottlenose Dol...	2020-02-10	2.70	180.00	14	4	4
8	8	Fluke	Young Botteno...	2023-08-25	1.90	95.00	14	4	4
9	9	Rocky	Brown Bear	2015-09-12	1.30	300.00	21	8	5

AnimalID	DietID	StartDT	EndDT	FeedTM
1	1	1	NULL	NULL
2	1	2	NULL	NULL
3	2	1	NULL	NULL
4	3	2	NULL	NULL
5	4	3	NULL	NULL
6	5	1	NULL	NULL
7	6	1	NULL	NULL
8	7	14	NULL	NULL
9	8	14	NULL	NULL
10	9	3	NULL	NULL
11	10	3	NULL	NULL
12	11	2	NULL	NULL
13	12	2	NULL	NULL
14	13	16	NULL	NULL
15	14	16	NULL	NULL

Results

Messages

	AnimalID	AName	Species	Habitat	Location	DietID	Diet
31	30	Tori	Bengal Tiger	Click to select the whole column	North Enclosure	1	Carnivore Diet
32	31	Cleo	African Lion		West Enclosure	1	Carnivore Diet
33	32	Toby	American Alligator	Rainforest	Aquatics Bay	1	Carnivore Diet
34	33	Sally	American Alligator	Rainforest	West Enclosure	1	Carnivore Diet
35	34	Theo	Cheetah	Grassland	North Enclosure	1	Carnivore Diet
36	35	Luna	Cheetah	Grassland	East Enclosure	1	Carnivore Diet
37	36	Rex	Giraffe	Grassland	South Enclosure	2	Herbivore Diet
38	37	Sasha	Giraffe	Grassland	West Enclosure	2	Herbivore Diet
39	38	Ace	African Elephant	Savanna	Aquatics Bay	2	Herbivore Diet
40	39	Olive	African Elephant	Savanna	South Enclosure	2	Herbivore Diet
41	40	Spike	Zebra	Grassland	Aquatics Bay	2	Herbivore Diet
42	41	Koda	Brown Bear	Forest	North Enclosure	3	Omnivore Diet
43	42	Raven	Brown Bear	Forest	South Enclosure	3	Omnivore Diet
44	43	Toby	Polar Bear	Arctic Tundra	East Enclosure	1	Carnivore Diet
45	43	Toby	Polar Bear	Arctic Tundra	East Enclosure	14	Marine Mammal Diet
46	44	Nina	Polar Bear	Arctic Tundra	West Enclosure	1	Carnivore Diet
47	44	Nina	Polar Bear	Arctic Tundra	West Enclosure	14	Marine Mammal Diet
48	45	Blake	Giraffe	Grassland	South Enclosure	2	Herbivore Diet
49	46	Sophie	Giraffe	Grassland	North Enclosure	2	Herbivore Diet
50	47	Mia	Bengal Tiger	Rainforest	East Enclosure	1	Carnivore Diet
51	48	Ace	Bengal Tiger	Rainforest	North Enclosure	1	Carnivore Diet
52	49	Hunter	African Lion	Savanna	Aquatics Bay	1	Carnivore Diet
53	50	Luna	African Lion	Savanna	West Enclosure	1	Carnivore Diet
54	51	Theo	Cheetah	Grassland	South Enclosure	1	Carnivore Diet
55	52	Jasmi	Cheetah	Grassland	East Enclosure	1	Carnivore Diet
56	53	Eliza	Red Kangaroo	Forest	Aquatics Bay	6	Frugivore Diet
57	54	Jack	Red Kangaroo	Forest	South Enclosure	6	Frugivore Diet
58	55	Max	Zebra	Grassland	East Enclosure	2	Herbivore Diet
59	55	Max	Zebra	Grassland	East Enclosure	24	Desert Herbivore ...
60	56	Zara	Zebra	Grassland	West Enclosure	2	Herbivore Diet
61	56	Zara	Zebra	Grassland	West Enclosure	24	Desert Herbivore ...
62	57	Nash	Koala	Forest	East Enclosure	9	Folivore Diet
63	58	Bella	Koala	Forest	South Enclosure	9	Folivore Diet
64	59	Gizmo	Orca Whale	Ocean	North Enclosure	14	Marine Mammal Diet
65	60	Harley	Orca Whale	Ocean	South Enclosure	14	Marine Mammal Diet
66	61	Harvey	American Alligator	Rainforest	West Enclosure	12	Reptile Carnivore ...
67	62	Gracie	American Alligator	Rainforest	Aquatics Bay	12	Reptile Carnivore ...
68	63	Theo	Giraffe	Grassland	North Enclosure	2	Herbivore Diet
69	64	Lena	Giraffe	Grassland	East Enclosure	2	Herbivore Diet
70	65	Rocky	Gray Wolf	Grassland	South Enclosure	1	Carnivore Diet
71	66	Mio	Gray Wolf	Grassland	West Enclosure	1	Carnivore Diet
72	67	Boris	Brown Bear	Forest	South Enclosure	1	Carnivore Diet
73	68	Tina	Brown Bear	Forest	North Enclosure	1	Carnivore Diet

